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LOG-PERIODIC FITTING APPLIED TO OPTION PRICING USING MONTE CARLO METHODS

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Log-Periodic Fitting Applied to Option Pricing Using Monte Carlo Methods

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This work is dedicated to those born on the outskirts of society who chose to defy the statistics.

Acknowledgments

This work is dedicated to all the people who devoted a little of their lives so that I could build the whole of mine. To my mother, Irene Alves de Jesus, the most extraordinary woman in the world, who fought with all her strength to make this dream possible. To my late father, José Miranda, a provider who never allowed us to go without what we needed. To my family; to my brothers, more than friends, Paulo Henrique and José Lucas, for our many years of togetherness. To my companions along the way, João Augusto, Ana Caroline Manso, Gracielle Arvelos, and Henrique Lettieri, for all the support they gave me. To all the friends and colleagues who shared moments of life with me. To Professor Leonardo Luíz e Castro, for his invaluable guidance and for teaching me values I will carry throughout my life. To Clarissa Martins, for believing in my dream and supporting me at the most difficult moment of my life: I dedicate much more than the joy of having completed this work; I dedicate the happiness of realizing that the difficult days we lived through were necessary to forge my character.

To all of you, thank you very much.

Time is fluid, and we are too stony
to understand that life changes.
— Clarissa Martins

Abstract

This work seeks to study stock markets in specific situations involving financial bubbles, investigating how derivatives linked to underlying assets tend to behave amid dramatic price fluctuations. These economic dynamics are studied using methods from statistical physics and techniques specific to econophysics, including the solution of differential equations and numerical and computational methods. Market dynamics are examined in relation to log-periodic behavior, which is used to model various natural and human phenomena.

Keywords: Econophysics; options; financial bubbles; economic crises; Monte Carlo; stock market; log-periodicity; Black–Scholes. **Translation note.** This document is an

English rendition of the undergraduate monograph submitted in Portuguese in 2019. Mathematical expressions, experimental values, figure numbering, references, and the research's historical context are preserved wherever possible. Charts are reproduced from the original thesis; some labels embedded within those original charts remain in Portuguese. The Portuguese original is the authoritative academic record. Minor typographic or mathematical inconsistencies in the source are flagged rather than silently amended.

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Introduction

The study of the dynamics of systems governed by economic behavior has attracted physicists throughout the world over recent decades. Despite the complexity of the movements observed in these systems—generally attributable to the many economic agents involved—concepts and techniques from statistical physics, together with numerical and computational procedures, offer ways to better understand their dynamics. This study examines a particular economic system characterized by two stages: a rapid increase in stock prices followed by an abrupt decline. Such an increase is called a bubble and is defined by a notable and substantial difference between the price of an asset and its underlying value. The definition itself is problematic because it involves the concept of an asset's value. We therefore focus mainly on quantitative results, without delving too deeply into conceptual questions.

History contains several well-known episodes of price bubbles. Two noteworthy events are the tulip mania of the seventeenth century and the Internet bubble of the 1990s, the latter characterized by speculation and rising share prices of Internet-related technology companies (the dot-com bubble). Economic crises are dramatic events and, as the global financial crisis of 2008 showed, may have devastating consequences for economic well-being across the world. One possible way to protect against dramatic variations in asset prices is to invest in options, futures contracts, and other financial derivatives. Options, in particular, can offer economic agents a way to obtain some protection for their assets and can be traded readily, just like shares on the market.

This work investigates how option exercise prices tend to diverge from the values of their underlying shares when a financial bubble occurs. These are exceptional moments in a stock's history, and established option-pricing methods were not developed specifically to correct such large differences. Under these conditions, there can be extreme gains and losses. We employ log-periodic fitting to model crashes in a set of indices and then examine option prices relative to the value of the underlying assets at those times. We also analyze fluctuations in stock prices during bubbles and crashes by studying logarithmic returns. For option pricing, we consider the Black–Scholes method, which derives option prices through the solution of a differential equation, and Monte Carlo methods, based on computer simulation. Finally, the work proposes a type of "solution" for options in bubble markets: a correction to the Monte Carlo method that takes account of log-periodic fits.

Part I

A Study of Market Dynamics

Chapter 1

A Brief Discussion of Derivatives and Physics Applied to Financial Markets

Over the years, financial market dynamics have become increasingly complex, drawing the attention of professionals who do not necessarily have formal training in economics or finance. We seek occurrences exhibiting regularities or patterns of behavior that can be systematically described with mathematical models. Given the complexity of systems in which the variables are largely governed by human behavior, our focus is restricted to quantitative analyses that may provide answers concerning the cases under examination.

A class of instruments particularly relevant to this objective is financial derivatives. A derivative is a financial instrument whose price depends on one or more variables associated with other market instruments—in our case, shares traded on stock exchanges. Although many types of derivatives exist, we concentrate on options, whose prices depend on those of their underlying assets.

1.1 Options

Options are a type of financial derivative, and their prices (premiums) derive from the value of the underlying asset. In the stock market, an option can be likened to an insurance policy or a deposit paid to secure the right to purchase an asset. Options have been traded on exchanges since 1973 and have grown substantially worldwide. They now exist for many kinds of assets, including stocks, currencies, commodities, and futures contracts. An option gives its buyer a right, but not an obligation, to buy or sell an asset at a specified price on or by a future date. The writer of the option, in contrast, assumes the corresponding contractual obligation. Options can be classified into two basic types. A *call* option gives its holder the right to buy the underlying asset on a specified date at a predetermined price. A *put* option gives its holder the right to sell the underlying asset on a specified date at a predetermined price.

Because the buyer is not required to exercise the contract at expiration, the options market is particularly dynamic and attractive to two groups: hedgers and speculators.

Hedgers seek to minimize losses and use options to reduce risks. For instance, someone planning to purchase a quantity X of US dollars on a specified future date may buy a call option to guard against appreciation of the dollar; this ensures the right to buy that amount at the agreed exchange rate. Speculators, by contrast, seek profits by anticipating rising or falling asset prices. If we buy a call option today with an exercise price of 50 monetary units and the share subsequently rises to 60, the option permits us to buy an asset for less than its market price. If the share falls below 50, exercising the option becomes unattractive and it expires worthless. Speculators attempt to anticipate these changes. In short, a call buyer expects the underlying price to rise, whereas a put buyer expects it to fall.

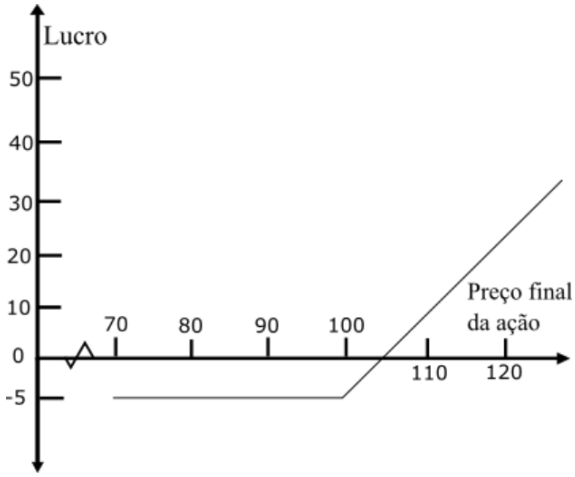


Figure 01. Profit from purchasing a European call option; premium 5 and strike price 100.

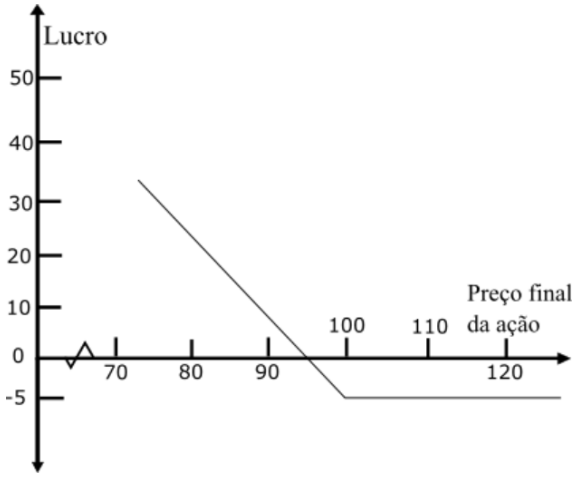


Figure 02. Profit from purchasing a European put option; premium 5 and strike price 100.

The other side of the option contract involves the economic agent writing the option. The profit profile for the writer differs substantially from that of the buyer. We wish to emphasize the “asymmetry” of option returns when the underlying asset undergoes a pronounced increase or decrease in value.

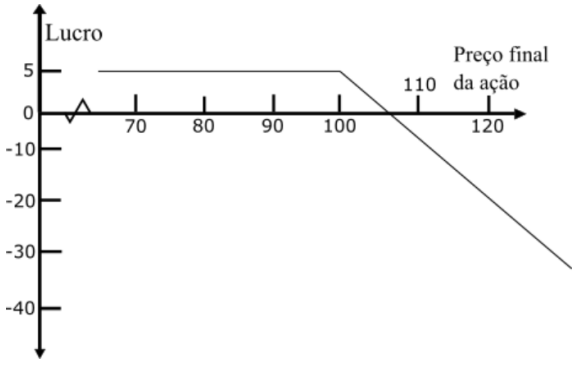


Figure 03. Profit from writing a European call option; premium 5 and strike price 100 (caption as in the original).

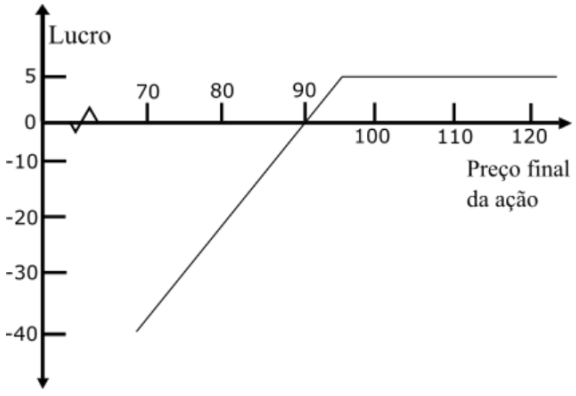


Figure 04. Profit from writing a European call option; premium 5 and strike price 95 (caption as in the original).

One motivation for this work is the observation that a financial crisis disrupts what

might otherwise be regarded as a “fair game,” in which both the buyer and the writer face risks of gains and losses. During a crisis, the burden may fall disproportionately on one side of the transaction.

In Figure 4, the writer of a European put option is described as facing increasingly large losses as the underlying share depreciates. In a normal economic environment, these risks are assumed by the writer; however, a financial crisis creates a different setting, since prices may continue to decline and losses that had previously seemed limited can persist, at least throughout the crisis. Comparing Figures 2 and 4 suggests a substantial difference between the losses borne by the parties. A buyer’s maximum loss is the option premium, whereas the writer’s losses may rise as the underlying falls. [1]

Consider a simple example. Suppose a stock’s price has risen to a hypothetical value of 100 as a result of a speculative bubble; the stock’s market price therefore differs from its underlying value. Let 100 be the peak price the stock will reach. European put options are then offered with a strike price of 110, allowing their holders to sell the underlying stock for 110 on the expiration date. If the bubble subsequently collapses and the stock rapidly falls to 65, the result is highly favorable for put buyers but requires the writers to bear unexpectedly large losses. We do not address the broader economic consequences of this example. Nevertheless, substantial profits during a bubble may further encourage speculative price increases: a bubble can, in a sense, feed on economic agents’ desire for ever-greater profits.

1.2 Financial Crises and Log-Periodicity

Financial crises have a long history and are fascinating events for professionals and researchers from many fields. Like natural disasters, epidemics, widespread traffic congestion, and other human phenomena, financial crises can have tragic consequences. Attempts to explain the causes of market crashes often lead to inconclusive answers; nevertheless, the unpredictability of these systems makes them compelling examples of complex systems. Roubini and Mihm (2010) describe how, as early as 2006, Professor Nouriel Roubini issued stark warnings about the future of the US economy. A speculative bubble had formed, while many economists maintained that housing prices reflected genuine growth rather than a bubble. Roubini described an even more severe possibility: if homeowners began to default on their loans, the global financial system itself could cease to function properly. Several aspects of his account subsequently materialized. This suggests that financial crises may exhibit recurring features that make partial anticipation possible. A crisis often follows a similar narrative: allegedly safe investments, assets said never to lose value, and promises of guaranteed profit. Speculation is also frequently accompanied by easy access to credit and low interest rates, further encouraging investment in supposedly risk-free opportunities.

[1] The source describes the payoff as that of a written put, although the original caption for Figure 4 calls it a written call. This textual inconsistency is retained here.

Important studies suggest that such events may exhibit similar properties as they approach extreme states. Here, our principal interest is the occurrence of log-periodicity (Sornette, 2003; Juliao et al., 2008).

$$y = \cos(\ln x) \tag{1.1}$$

In his investigation of log-periodic behavior, Sornette observed that structures of this type appear in various phenomena associated with a rupture, as though the mathematical expression captured a collective response of the agents involved. Broadly speaking, a market crash involves large numbers of participants submitting sell orders at nearly the same time. These investors generally do not know one another and have not coordinated their sales, yet their joint actions can sharply reduce an asset's price. According to Sornette, a model of stock-market crashes should take into account the following characteristics:

- investors are influenced by their neighbors;
- local imitation can propagate spontaneously into global cooperation;
- global coordination among investors may precipitate a crash;
- prices are linked to the system's properties;
- the system's parameters evolve slowly over time.

These concepts motivate our use of log-periodic fitting, discussed below.

Chapter 2

Stocks and Indices Under Study

The analyses in this chapter draw on examples in Chatterjee and Chakrabarti (2006). We select four market series: one asset and three equity indices—Bitcoin in US dollars (BTC-USD), the Nikkei 225 (N225), the S&P 500 (GSPC), and the NASDAQ (IXIC).

Why study indices? Indices provide a general average of the behavior of their constituent shares. Although the index itself is not necessarily traded as an individual asset, its aggregate representation can be used to understand a sector or market and to construct simulations. Bitcoin was selected because its price exhibited an especially striking bubble and decline within a comparatively short period. Table 1 lists the chosen series and the historical intervals identified as bubble and crisis periods.

Table 2.1: Bubble and crisis periods for the studied series (dates retained from the original).

Series	Bubble period	Crisis period
Bitcoin	09/07/2017–12/16/2017	12/16/2017–04/10/2018
Nikkei 225	01/15/1984–12/24/1989	12/31/1989–08/29/2004
NASDAQ	06/01/1995–03/26/2000	04/02/2000–03/30/2003
S&P 500	03/01/2004–10/07/2007	10/14/2007–05/31/2009

The following figures illustrate these periods and the dramatic increases and declines in asset values. Additional information can be extracted from the normalized absolute logarithmic returns.

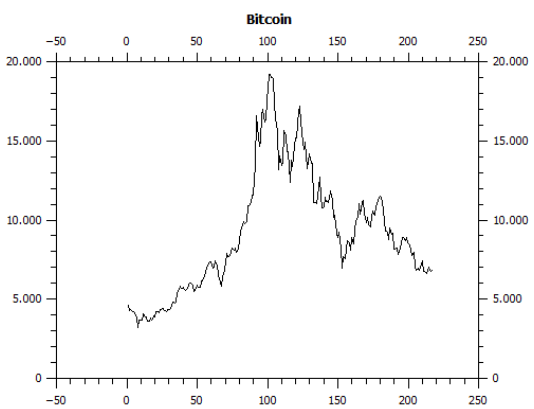


Figure 05. Bitcoin price history.

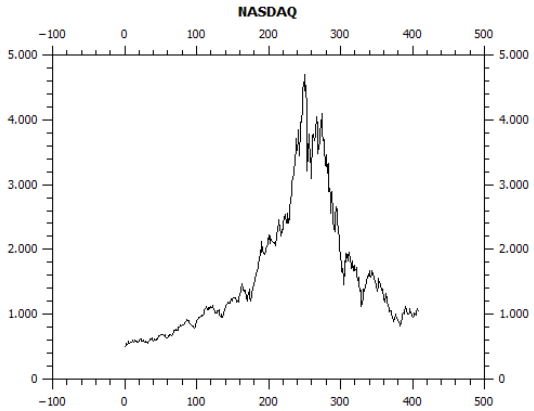


Figure 06. NASDAQ price history.

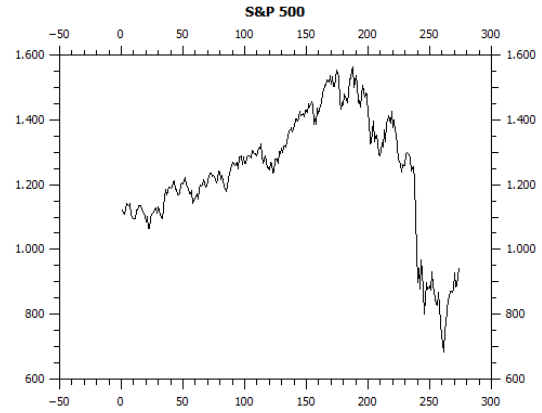


Figure 07. S&P 500 price history.

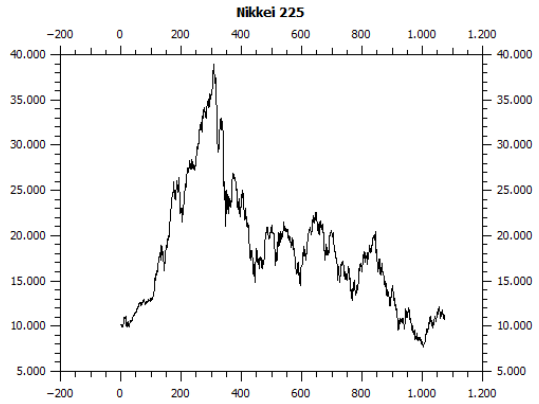


Figure 08. Nikkei 225 price history.

We examine the probability that the normalized absolute log return exceeds a value x, written $P(V > x)$. The aim is to assess the incidence of large absolute returns in the bubble and crisis periods of the studied series. A stock return is a measure of the price variation over a given time interval.

If time is divided into discrete intervals (days, weeks, or months), the absolute logarithmic return is

$$|R(t)| = |\ln S(t) - \ln S(t-1)| \tag{2.1}$$

where $S(t)$ is the stock price at time t. We define the normalized quantity

$$V(t) = \frac{|R(t)|}{\sigma} \tag{2.2}$$

where σ is the standard deviation of $|R(t)|$. We then plot $P(V(t) > x)$ as a function of x on semilogarithmic axes.

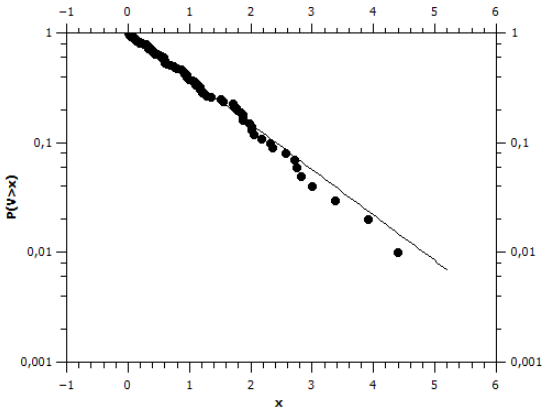


Figure 09. Exceedance probabilities for Bitcoin during the bubble.

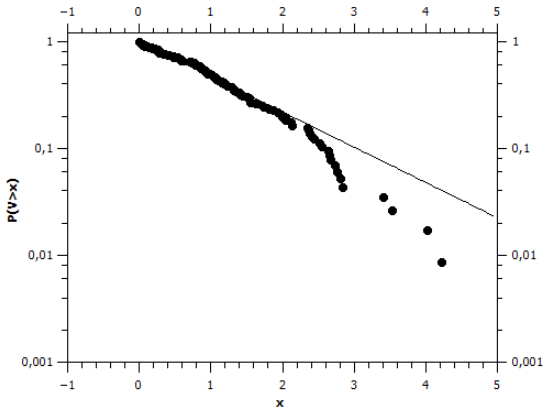


Figure 10. Exceedance probabilities for Bitcoin during the crash.

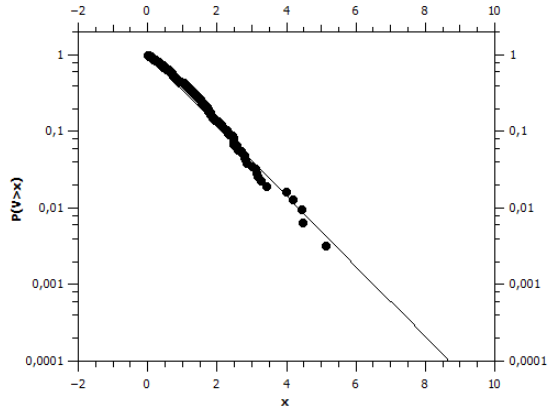


Figure 11. Exceedance probabilities for the Nikkei 225 during the bubble.

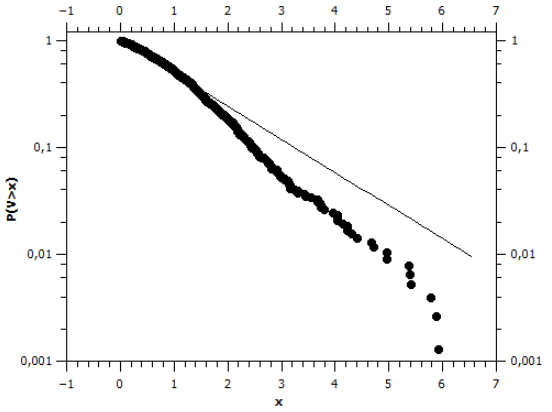


Figure 12. Exceedance probabilities for the Nikkei 225 during the crash.

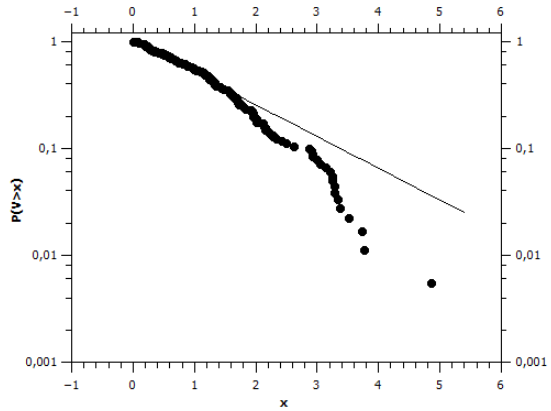


Figure 13. Exceedance probabilities for the S&P 500 during the bubble.

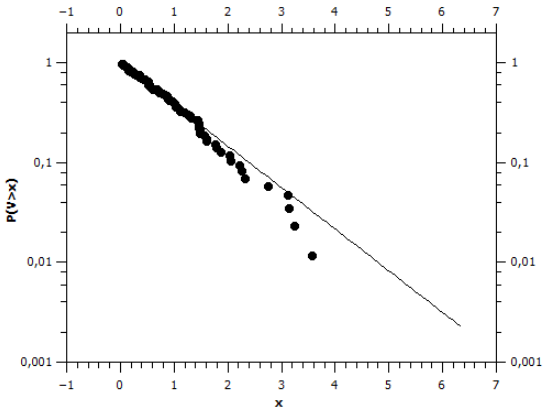


Figure 14. Exceedance probabilities for the S&P 500 during the crash.

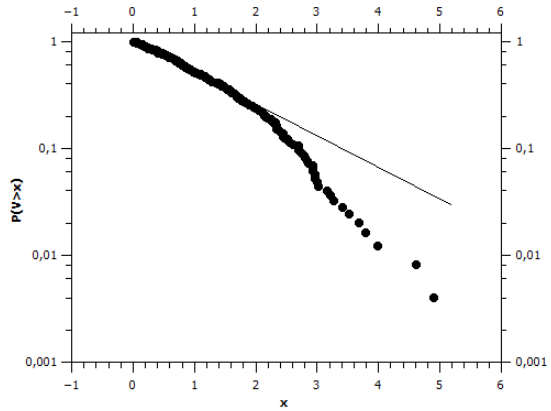


Figure 15. Exceedance probabilities for NASDAQ during the bubble.

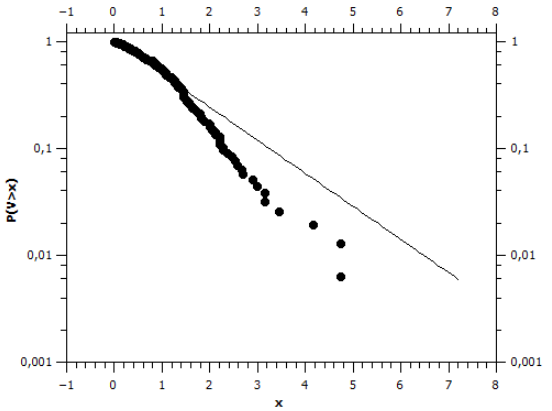


Figure 16. Exceedance probabilities for NASDAQ during the crash.

The curves shown in Figures 9–16 are fitted by

$$P(V > x) = \exp(-ax). \tag{2.3}$$

The values of the fitting coefficient a are reported in Table 2.

Table 2.2: Fit coefficients for Figures 9–16.

Series	Bubble period	Crisis period
Bitcoin	0.95	0.76
Nikkei 225	1.05	0.71
S&P 500	0.68	0.96
NASDAQ	0.67	0.71

Using Bitcoin as an example, the fitting coefficient is larger for the bubble period than for the crash period. Consequently, large absolute log returns are more likely during the crash: for Bitcoin, price changes were more severe in the crisis phase. Figure 17 illustrates this contrast, and analogous comparisons can be made for the other indices.

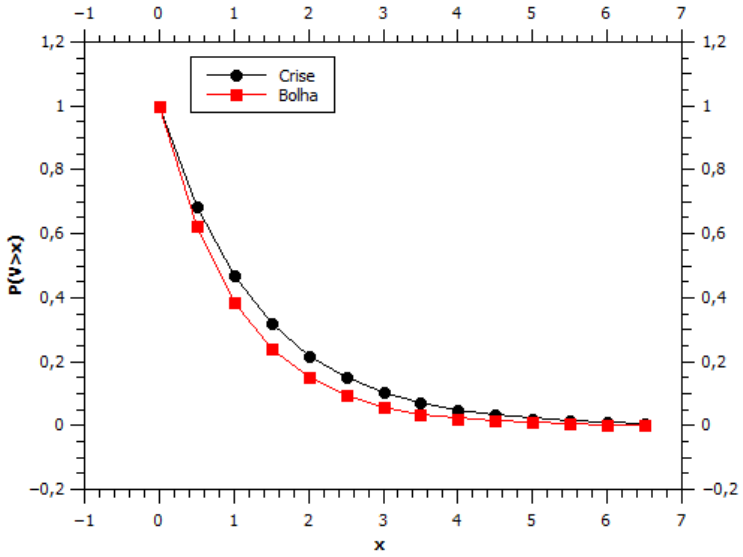


Figure 2.1: Comparison of the distributions $P(V > x)$ for Bitcoin during the bubble and crash periods.

At the beginning of the log-return analysis, we anticipated finding a consistent relationship between the fitting parameters in bubble and crisis periods. Larger values of a correspond to faster-decaying curves, and thus smaller probabilities of exceeding a specified threshold x. Such a pattern would have allowed a simpler model, but it did not emerge. Chatterjee and Chakrabarti (2006) discuss the possibility that the exponential fit $\exp(-ax)$ may not always be appropriate and that a power-law fit x^{-a} may be more suitable in bubble periods.

Crisis periods create dramatic conditions that can harm broad sectors of the economy. The S&P 500 series is associated with the 2008 financial crisis, rooted in a speculative housing bubble. That crisis affected the entire world and left lasting consequences. Because avoiding such crises is important, researchers seek models capable of identifying risks and anticipating market disruptions. We now examine one such model: the

log-periodic fit.

Chapter 3

The Log-Periodic Fit

Building on the preceding discussion of log-periodic behavior, we fit the financial time series using a log-periodic power law:

$$y(t) = A + B(t_c - t)^z + C(t_c - t)^z \cos\Big(\omega \log(t_c - t) + \Phi\Big). \tag{3.1}$$

The parameter t_c represents the most likely time of the crash, subject to uncertainty. The parameter z controls the power-law growth, while ω governs the oscillation frequency in logarithmic time; the coefficients A, B, C, and Φ determine the shape of the fitted curve and are not assigned separate explicit physical meanings in this study. The log-periodic power law is used in multiple settings, including the analysis of potential financial crashes (Jacobsson, 2009). Here, we fit this expression to the time evolution of the selected assets.

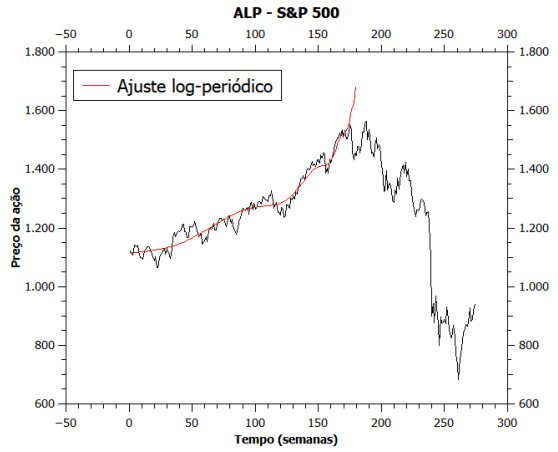


Figure 18. Log-periodic fit for the S&P 500 index.

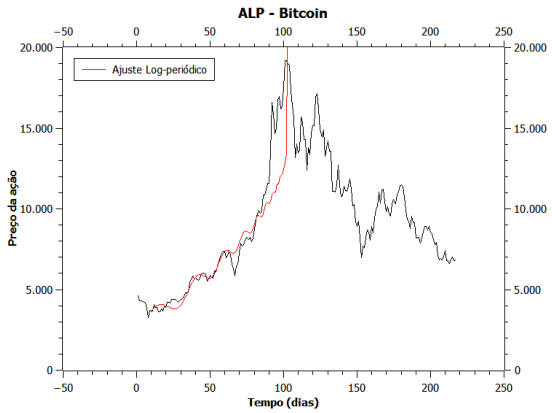


Figure 19. Log-periodic fit for Bitcoin.

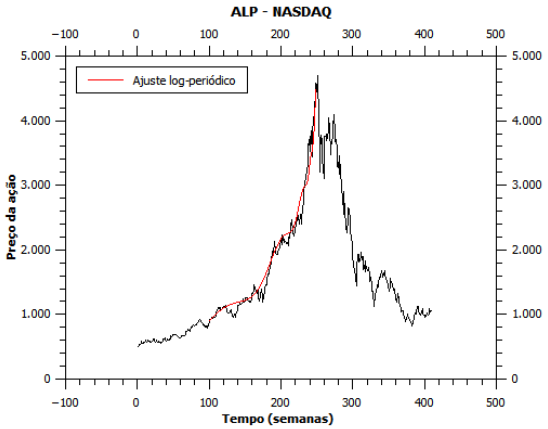


Figure 20. Log-periodic fit for NASDAQ.

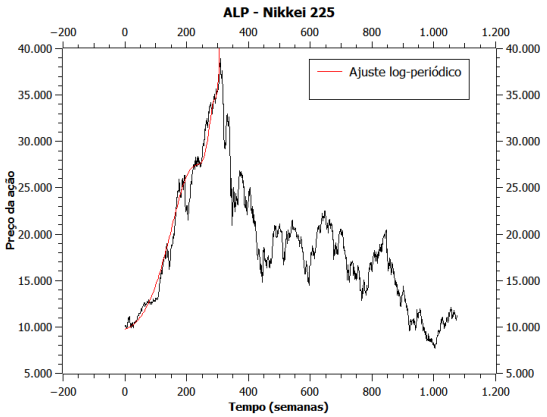


Figure 21. Log-periodic fit for the Nikkei 225.

The fitted parameters are given in Table 3.

Table 3.1: Log-periodic fit parameters for the studied series. Values follow the original thesis, with decimal punctuation normalized.

Parameter	Bitcoin	NASDAQ	Nikkei 225	S&P 500
A	13,865	4,806	39,200	2,177
B	-818.5	-722.9	-897	-425.8
C	46.8	31.3	118.4	9.08
t_c	102.5	248.4	307.9	182.3
Φ	-1.2	5.6	-6.3	1.14
z	0.568	0.345	0.6	0.17
ω	39.3	-14.4	9.3	15.3

Throughout this study, the fit worked well for most indices, with some difficulties arising for Bitcoin's historical data. Bitcoin exhibited unusually rapid appreciation, to such an extent that the fitted log-periodic curve could not fully anticipate or reproduce the price increase. As discussed in Chapter 1, the model assumes that system parameters change slowly over time, whereas Bitcoin changed drastically within a short period.

Overall, the fitting procedure proved useful, and the resulting parameters are employed later in computational simulations. We next discuss how these curves will enter our option-pricing analysis.

3.0.1 Applying the Log-Periodic Fit to Option Pricing

The preceding results suggest that log-periodic fitting may be used in several applications, including analysis of the timing of a market crash. We intend to combine the fitted curve with a numerical pricing technique introduced later. The idea is to model a market whose prices remain free to fluctuate but nevertheless tend to follow a log-periodic trend. The fitted log-periodic function therefore acts as a weighting influence on the time evolution of asset prices. Although prices can fluctuate randomly, their paths tend toward the fitted trajectory. Before presenting the numerical procedures, we review the best-known analytical option-pricing approach, obtained from a stochastic differential

equation.

Part II

Option-Pricing Methods

Chapter 4

The Black–Scholes Method

The Black–Scholes formula is among the most widely used methods for pricing options. Because it provides an explicit analytical solution, it can be adapted to many applications. Numerous software packages already implement this formula, often making numerical procedures such as Monte Carlo simulation unnecessary for the simplest cases. Stock-price behavior is commonly modeled with Wiener processes, known in physics as Brownian motion. A Wiener process is a particular type of Markov stochastic process. A process is Markovian when only the current state of a variable is relevant for predicting its future evolution: its previous history need not be known (Vasconcelos, 2004; Hull, 2016).

Since the option's price is a function of the underlying share price and time, we can treat it as a function of stochastic variables. One important result in stochastic calculus is Itô's lemma. Suppose a variable x follows an Itô process,

$$dx = a(x,t)\,dt + b(x,t)\,dz, \tag{4.1}$$

where dz denotes the increment of a Wiener process. Itô's lemma states that a function $G(x,t)$ evolves according to

$$dG = \left(\frac{\partial G}{\partial x}a + \frac{\partial G}{\partial t} + \frac{1}{2}\frac{\partial^2 G}{\partial x^2}b^2\right)dt + \frac{\partial G}{\partial x}b\,dz. \tag{4.2}$$

We are interested not in an arbitrary variable x but in S, the underlying stock price. Equation (4.1) then becomes

$$dS = \mu S\,dt + \sigma S\,dz, \tag{4.3}$$

where μ is the drift rate and σ is the asset's historical volatility. Combining Eqs. (4.2) and (4.3) gives

$$df = \left(\frac{\partial f}{\partial S}\mu S + \frac{\partial f}{\partial t} + \frac{1}{2}\frac{\partial^2 f}{\partial S^2}\sigma^2 S^2\right)dt + \frac{\partial f}{\partial S}\sigma S\,dz. \tag{4.4}$$

The next step is to choose a portfolio combining the share and its derivative in a way that eliminates the random risk generated by the Wiener process. We write the portfolio value as

$$\Pi = -f + \frac{\partial f}{\partial S}S. \tag{4.5}$$

Over a small time interval Δt, its change is

$$\Delta\Pi = -\Delta f + \frac{\partial f}{\partial S}\Delta S. \tag{4.6}$$

Substituting the discrete forms of Eqs. (4.3) and (4.4) into Eq. (4.6) yields

$$\Delta\Pi = -\left(\frac{\partial f}{\partial t} + \frac{1}{2}\frac{\partial^2 f}{\partial S^2}\sigma^2 S^2\right)\Delta t. \tag{4.7}$$

The Wiener increment Δz has disappeared from this equation, so the chosen portfolio is locally riskless during the interval Δt. Assuming that an arbitrage opportunity cannot provide a risk-free return above the market's risk-free interest rate, we obtain

$$\Delta\Pi = r\Pi\Delta t, \tag{4.8}$$

where r is the risk-free interest rate. Substituting Eqs. (4.5) and (4.7) into Eq. (4.8) gives

$$-\left(\frac{\partial f}{\partial t} + \frac{1}{2}\sigma^2 S^2\frac{\partial^2 f}{\partial S^2}\right)\Delta t = r\left(-f + \frac{\partial f}{\partial S}S\right)\Delta t. \tag{4.9}$$

Rearranging the terms, we reach

$$\frac{\partial f}{\partial t} + rS\frac{\partial f}{\partial S} + \frac{1}{2}\sigma^2 S^2\frac{\partial^2 f}{\partial S^2} = rf. \tag{4.10}$$

This is the Black–Scholes differential equation. It admits different solutions according to the boundary conditions, which determine the derivative being priced. In this work, we primarily consider European put options. Call prices can be obtained through put–call parity. The formula for a European put is

$$p = Xe^{-r(T-t)}N(-d_2) - SN(-d_1), \tag{4.11}$$

where $N(x)$ is the cumulative distribution function of a standard normal random variable, with mean zero and unit standard deviation. The parameters appearing in the formula are

$$d_1 = \frac{\ln(S/X) + (r + \sigma^2/2)(T-t)}{\sigma\sqrt{T-t}}, \tag{4.12}$$

$$d_2 = \frac{\ln(S/X) + (r - \sigma^2/2)(T-t)}{\sigma\sqrt{T-t}}. \tag{4.13}$$

Editorial note: In the source, Equation (4.13) was also labeled d_1, although the accompanying text introduces both d_1 and d_2. Here the second label is rendered as d_2 to match the formula in Eq. (4.11); the functional expression itself is unchanged.

Numerically computing the cumulative normal distribution in Eq. (4.11) requires either tabulated values or an appropriate numerical method. The volatility σ is estimated from historical stock prices. Most modern software environments provide built-in im-

plementations of option-pricing formulae; nevertheless, the explicit solution does not meet the particular goal of this project. We therefore turn to numerical methods (Dymochkin et al., 2018). Our objective is to adapt a familiar computational approach so that option prices may reflect an underlying asset following a log-periodic trend.

Chapter 5

Option Pricing by the Monte Carlo Method

As computing technology has advanced, numerical methods have become increasingly important in financial calculations, owing to improvements in their speed and efficiency. A major advantage is the flexibility to modify and extend computational procedures. Monte Carlo simulation is particularly suitable for the option-pricing problem studied here. It works by simulating many hypothetical price paths and the resulting option payoffs, with the stochastic evolution following a random process also familiar from physics (Crack, 2009; Boyle, Broadie, and Glasserman, 1997).

We begin by presenting the method and explaining the pricing program. The law of large numbers, a fundamental result in probability theory, states that for independent observations $X_1, X_2, \ldots, X_n$ satisfying

$$E(X_i) = \mu, \tag{5.1}$$

with finite μ, their sample mean tends toward μ as the number of observations increases:

$$\overline{X}_n \equiv \frac{1}{n}\sum_{i=1}^{n} X_i \longrightarrow \mu \qquad (n \to \infty). \tag{5.2}$$

The basic idea of our software is to generate many possible simulated values of the payoff $V(T)$ at the option's expiration date T. Randomness drives the simulations, with the aim of producing scenarios representative of possible real-world trajectories.

The fundamental risk-neutral valuation relation is

$$V(t) = e^{-r(T-t)} E_t^*[V(T)]. \tag{5.3}$$

It is the expected payoff, discounted at the risk-free rate, under a risk-neutral probability measure. Our random variable is the terminal payoff $V(T)$, whose expected value will be estimated numerically. A first simulated path produces $V(T)_1$; subsequent paths give

$V(T)_2$, $V(T)_3$, and so on until $V(T)_n$. The average is

$$\overline{V} = \frac{1}{n}\sum_{i=1}^{n} V(T)_i, \tag{5.4}$$

which tends to the expected payoff as n increases. The Monte Carlo option-price estimator is consequently

$$\widehat{V}(t) = e^{-r(T-t)}\overline{V} = \frac{e^{-r(T-t)}}{n}\sum_{i=1}^{n} V(T)_i. \tag{5.5}$$

We have not yet specified how the random samples $V(T)_i$ are generated. Just as the Black–Scholes model has a well-defined underlying dynamics, the Monte Carlo algorithm employs a stochastic update equation:

$$S(t_{j+1}) = S(t_j)\exp\left[\left(r - \frac{\sigma^2}{2}\right)\Delta t + \sigma\sqrt{\Delta t}\,\epsilon_{j+1}\right]. \tag{5.6}$$

We divide the time interval into equal steps, taking $\Delta t = 1/365$, and choose an initial stock price $S(t_0)$. Each trajectory leads to a terminal put payoff

$$V(T)_i = \max\left(\text{stk} - S(t_J)_i,\, 0\right), \tag{5.7}$$

where J is the number of days until expiration and stk is the strike price. As the number of simulated paths increases, the mean payoff converges to the value expected from Black–Scholes under the model assumptions. Testing the accuracy of the simulation against the analytical result is therefore an essential part of this study.

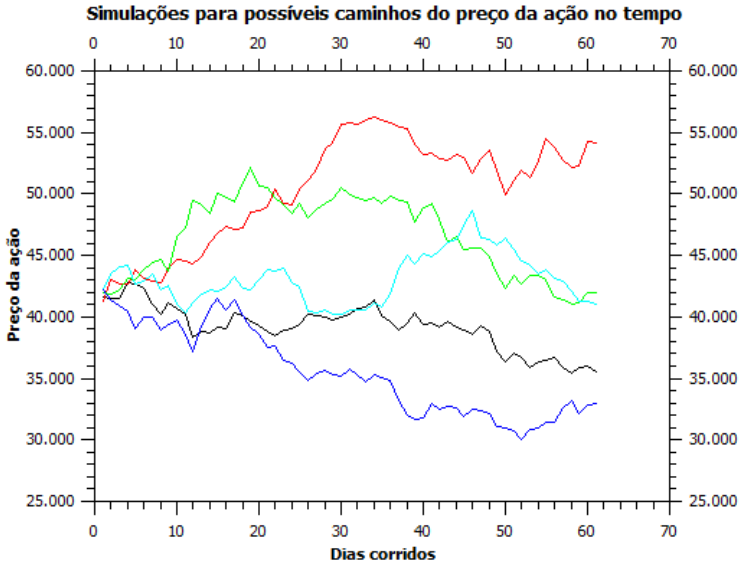


Figure 5.1: Examples of simulated trajectories of the underlying stock price over time. The program generates many such paths to estimate the final option value.

5.1 Simulation Algorithm

The program begins by reading the key simulation inputs: the strike price at expiration, the initial stock price, the interest rate, the asset's volatility, and the number of days remaining before expiration. These values are inserted into Eq. (5.6) while j ranges from 0 to $J-1$. At the end of the path, the simulation reaches $S(t_J)$, the modeled stock price at expiration. The corresponding payoff is computed using Eq. (5.7) and stored. The process is repeated n times; a larger number of paths generally improves the accuracy of the estimate. In the limit $n \to \infty$, the average approaches the expected option price.

5.2 Example: Pricing a European Put with Monte Carlo

Following an example from Crack (2009), we simulate an option with the input values in Table 4.

Table 5.1: Parameters for the European put valuation example.

Variable	Value
Strike λ	\$45.00
Initial price S_0	\$41.75
Volatility σ	0.34
Interest rate r	5.35%
Days to expiration J	61

Here λ is the strike, S_0 the starting price, and r the interest rate. Equation (5.6) becomes

$$S(t_{j+1}) = S(t_j) \exp\left[\left(0.0535 - \tfrac{1}{2}(0.34)^2\right)\frac{1}{365} + 0.34\sqrt{\frac{1}{365}}\,\epsilon_{j+1}\right]. \tag{5.8}$$

In this example $j = 0, 1, 2, \ldots, J-1$, while $\epsilon_1, \epsilon_2, \ldots, \epsilon_J$ are standard normal random values with mean zero and standard deviation one. As n increases, the mean payoff tends toward the Black–Scholes price of approximately \$4.08 for the European put.

We developed a program to assess how pricing accuracy improves as the number of paths increases. It computes successive estimates using progressively larger sample sizes. We observed that after roughly $n = 20{,}000$ simulations, additional increases in n produced more gradual improvements in convergence toward the expected option price. Figure 23 shows this behavior for n ranging from 0 to 100,000.

Three techniques were used to generate approximately normal random variables: a method based on the central limit theorem (CLT), one described by Abramowitz and Stegun, and a method described by G. Marsaglia and T. A. Bray. The particular choice did not noticeably alter the simulation's convergence, so subsequent analyses use the CLT-based generator because it was faster in these tests.

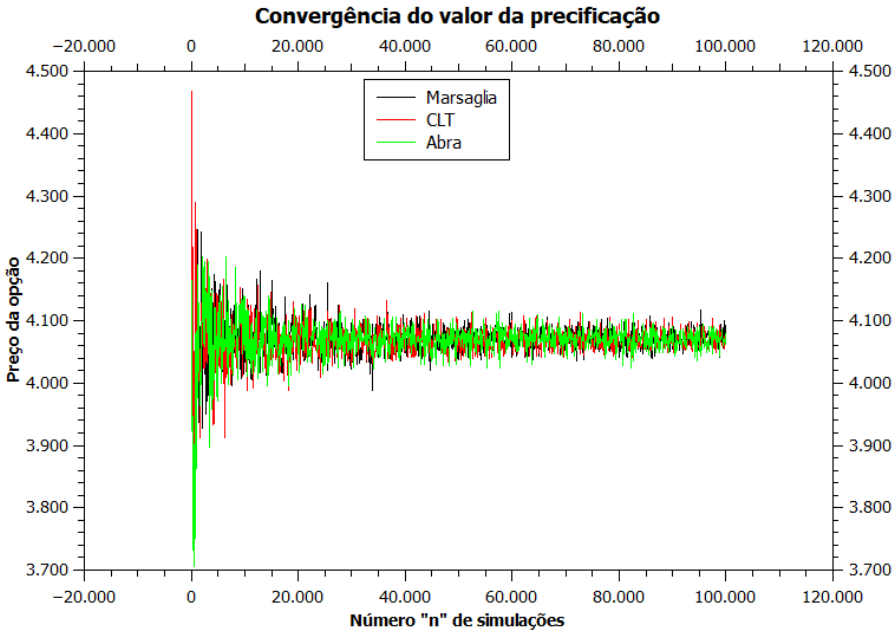


Figure 5.2: Convergence of the estimated option value with increasing simulation count, comparing three random-variable generation methods.

An important limitation of Monte Carlo pricing is computation time. Although suitable for the research conducted here, our program took approximately 55.8 seconds to calculate an option value. That duration may be acceptable in an academic study but is impractical for a highly dynamic stock market, where the underlying price may change substantially before the calculation finishes. We used $n = 2{,}500{,}000$ paths, a fairly large number that nonetheless gave satisfactory convergence toward \$4.08. Running 100 independent pricing experiments with this number of paths yielded the histogram in Figure 24. The estimates had a mean of \$4.082 and a reported error of 0.001:

$$\widehat{V}(T) = 4.082 \pm 0.001. \tag{5.9}$$

The software produced estimates with more decimal places, but for the pricing discussion we only need precision to two decimal places, corresponding to cents.

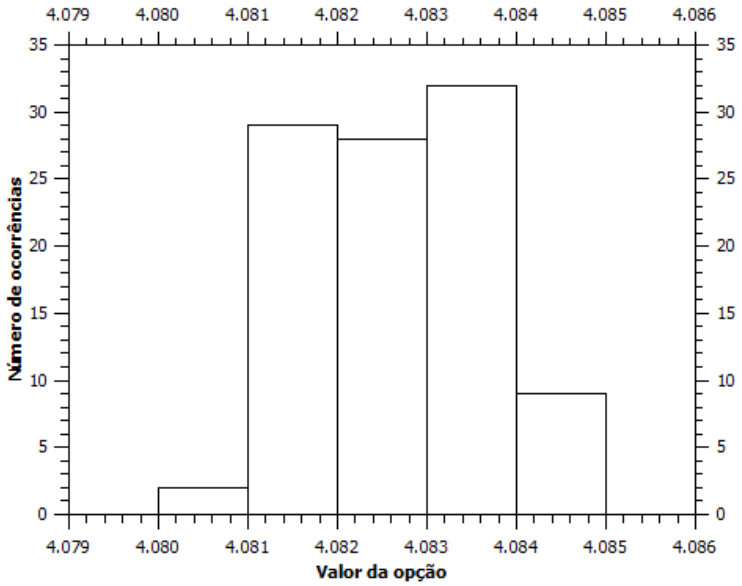


Figure 5.3: Histogram of option-value estimates and their frequencies for $n = 2{,}500{,}000$ paths in each simulation.

To further identify the range of n over which the program converges satisfactorily, we repeated the calculation with n varying from 0 to 1,465,000, as shown in Figure 25. Figure 26 provides a closer view. For n greater than approximately 1,300,000, convergence becomes clearer. Next we investigate a simple variance-reduction procedure, the antithetic-variable technique.

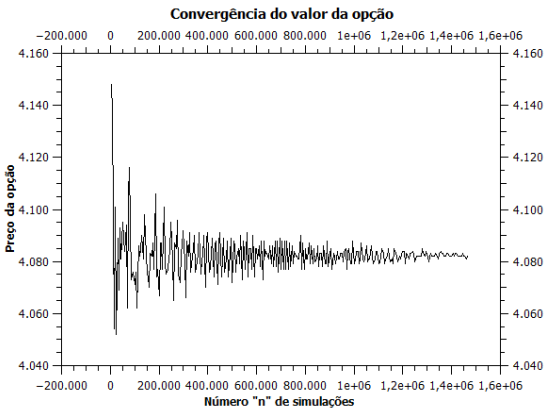


Figure 25. Convergence of the Monte Carlo estimate using the CLT method for n from 0 to 1,465,000.

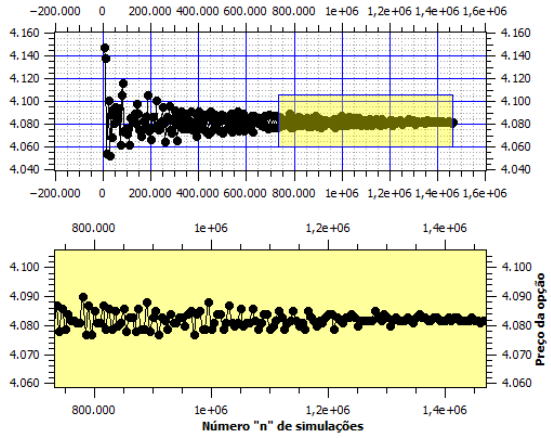


Figure 26. Enlarged view of the convergence plot in Figure 25.

5.2.1 Antithetic-Variable Technique

Several methods can reduce Monte Carlo variance and improve the precision of numerical estimates. The antithetic-variable technique introduces a simple change to the original program and can substantially reduce variability. In addition to simulating option payoffs using random numbers ϵ_j, the method also evaluates payoffs using their

negatives $-\epsilon_j$. The resulting estimator takes the average

$$\frac{\widehat{V}(T) + \widehat{V}'(T)}{2}. \tag{5.10}$$

This paired estimate is generally more precise, in the sense that its variance can be smaller than that of an unpaired estimator. We found that the extra computation also increased execution time. Comparing Figures 27 and 23 shows that the antithetic-variable technique improves precision for the same range of n. Other Monte Carlo variance-reduction methods are available but are not considered in this work.

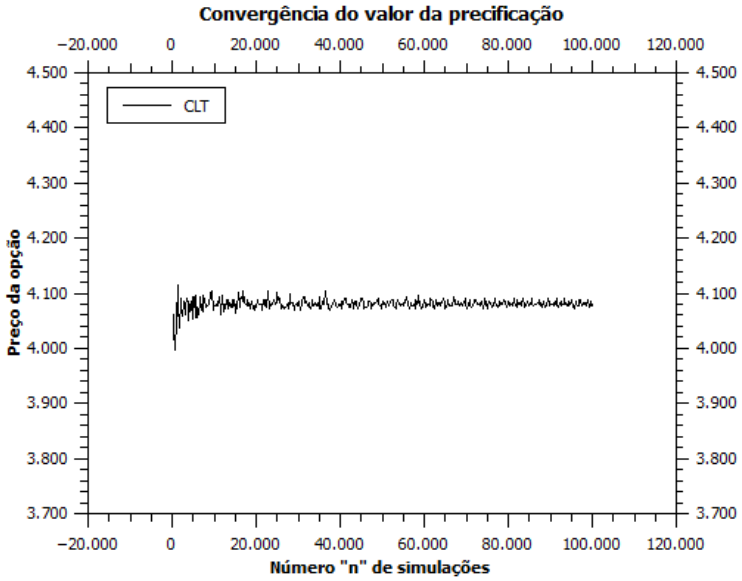


Figure 5.4: Convergence of estimated option prices with increasing sample size using antithetic variables.

5.3 Estimating Volatility from Historical Data

One of the principal quantities in this study is the volatility σ, which measures fluctuations in asset prices. We estimate it for the selected assets using their historical price series (Hirsa and Neftci, 2013; Crack, 2009). Let the inputs be $n+1$ price observations, S_i the asset price at step i, and τ the time interval measured in years.

Define the logarithmic return

$$u_i = \ln\left(\frac{S_i}{S_{i-1}}\right), \quad i = 1, 2, \ldots, n. \tag{5.11}$$

The sample standard deviation of the log returns is

$$s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(u_i - \overline{u})^2}, \tag{5.12}$$

where $\overline{u}$ denotes the sample mean. An equivalent computational form is

$$s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n} u_i^2 - \frac{1}{n(n-1)}\left(\sum_{i=1}^{n} u_i\right)^2}. \tag{5.13}$$

The historical annualized volatility is estimated by

$$s^* = \sigma = \frac{s}{\sqrt{\tau}}. \tag{5.14}$$

The original study gives the approximate standard error as $s^*/\sqrt{2n}$. We take $\tau = 1/252$, using 252 trading days in a year. We calculate historical volatilities at two moments during the bubble period for each of the four series: first before the price peak, and then near the peak, to investigate how this input affects option pricing as the market enters a crash.

Table 5.2: Historical volatility for two different intervals within each series' bubble phase.

Series	Interval 1	σ_1	Interval 2 / σ_2
Bitcoin	09/07/17–10/26/17	59.8%	09/07/17–12/14/17 / 62.52%
Nikkei 225	01/15/84–01/04/87	19.74%	01/15/84–12/10/89 / 20.96%
NASDAQ	06/01/95–10/19/97	29.26%	06/01/95–03/05/00 / 35.96%
S&P 500	03/01/04–12/18/05	14.17%	03/01/04–09/23/07 / 14.82%

Using these data, we next compare hypothetical option valuations with observed prices at later dates. The idea is to price an option using historical values at a given date, select a later expiration date, and then assess whether the resulting contract might have been profitable to its buyer.

5.4 Monte Carlo Applied to Historical Cases

We first analyze Bitcoin. On October 26, 2017, its value was reported as \$5,810, historical volatility was 59.8%, the chosen interest rate was 5.35%, and expiration was set 30 days ahead. We must also specify the strike, an assumed future price of the underlying asset. We consider the exceptional scenario in which the stock price rises by an extraordinary 25% in 30 days, from \$5,810 to \$7,262.50, and use the latter as the strike. With these inputs, the program estimated an option premium of \$1,467.60. Taking account of the option value and the underlying asset price, the original study discusses circumstances under which this contract could provide an economic advantage.

What happened in reality? On November 25, 2017, 30 days after the valuation date, Bitcoin traded at about \$8,766, considerably above the 25% increase assumed. Even allowing for uncertainty, the strike was well below the actual price at expiration. We also consider pricing an option just before the crash. On December 14, 2017, the Bitcoin closing price was reported as \$16,383 and historical volatility as 62.52%. Assuming

again a 5.35% interest rate, a strike of $20,478.75 (25% above the starting price), and an expiration horizon of 60 days, the model gave an option price of $4,429.13.

The actual subsequent movement was very different and favored holders of put options. During the crash, the Bitcoin price fell to $8,901. The holder of a put with a strike of $20,478.75 could sell an asset then worth $8,901 for the contractual exercise price, a difference of $11,577.75 that would be borne by the option writer. The values used in these illustrations are large compared with typical individual financial contracts, but they convey the central idea. In practice, options may also be traded in packages comprising multiple contracts, for example blocks of 100.

We applied similar comparisons to the other three indices, evaluating options first during the inflation of the bubble and then near the beginning of the crash. Results reported in the original study are reproduced in Tables 6 and 7. *The numerical values in the tables are preserved even where they do not fully match the separate narrative examples.*

Table 5.3: Option pricing for a date within the bubble-growth period (original Table 6).

	Bitcoin	Nikkei	NASDAQ	S&P 500
Valuation date	10/26/17	01/04/87	09/05/1999	09/24/2006
Price on valuation date	5,890	18,810	2,535	1,336
Volatility σ	59.80%	19.74%	52.86%	22.82%
Days to expiration	30	90	90	90
Strike price	6,183	19,738	2,870	1,412
Option premium	596.31	1,135.11	455.40	95.03
Underlying at expiration	8,766	22,739	3,203	1,418

Table 5.4: Option pricing for a valuation date close to the crash (original Table 7).

	Bitcoin	Nikkei	NASDAQ	S&P 500
Valuation date	12/14/17	12/10/89	03/05/2000	09/23/2007
Price on valuation date	16,383	38,271	4,587.16	1,526.75
Volatility σ	62.52%	20.96%	35.96%	14.82%
Days to expiration	30	90	90	90
Strike price	17,236	40,276	4,999.54	1,583
Option premium	1,617	2,474.25	542.03	65.55
Underlying at expiration	14,191	32,616	3,760.26	1,411.63

The source notes that the put option would be profitable to its holder in the post-peak cases shown. In the next chapter, we discuss why these observations motivate a correction to numerical option pricing. First, however, we briefly address the corresponding behavior of call options.

5.4.1 The Counterpart: Call Options

The preceding discussion concerned the advantages and disadvantages of holding put options during bubbles and crashes. How do call options behave in these periods? Broadly speaking, the pattern is reversed: call holders are favored during a rising bubble and disadvantaged during a crash. The same style of analysis can be applied to call options, so it is not repeated here.

Having completed the proposed analyses, we now apply the log-periodic fit to the numerical method developed in this chapter.

Part III

The Log-Periodic Fit Applied to Monte Carlo

Chapter 6

Log-Periodic Fitting Applied to Monte Carlo Option Pricing

We now consider the central analysis of this thesis: a modification of the Monte Carlo numerical method for option pricing. The advantage of a computational approach is that we can implement changes and quickly examine whether their behavior is consistent with our expectations.

Figure 22 shows that the standard Monte Carlo pricing algorithm allows stock prices to evolve randomly over time. This randomness is represented by ϵ_j in Eq. (5.6). Here, however, we investigate the hypothesis that prices during an economic bubble do not follow entirely random paths but have an overall tendency to rise. As discussed earlier, log-periodicity may provide a possible description of such growth. We therefore combine standard Monte Carlo simulation with the log-periodic fit, which serves as a model for possible movements of financial indices. Our proposed modification of Eq. (5.6) is

$$S(t_{j+1}) = \frac{1}{2}\Bigg\{S(t_j)\exp\left[\left(r-\frac{\sigma^2}{2}\right)\Delta t + \sigma\sqrt{\Delta t}\,\epsilon_{j+1}\right] + \frac{1}{2}L(t_{j+1})\left(1+e^{\sigma\epsilon_{j+1}}\right)\Bigg\}. \tag{6.1}$$

Here $L(t_{j+1})$ is the fitted log-periodic value at the discrete time t_{j+1}. The term $\frac{1}{2}L(t_{j+1})(1+e^{\sigma\epsilon_{j+1}})$ was introduced to represent the assumed behavior of a market during a bubble. It averages the deterministic value from the log-periodic curve and a randomly perturbed version of that value, allowing small deviations from the fitted trend. By writing

$$L'(t_{j+1}) = \frac{1}{2}L(t_{j+1})\left(1+e^{\sigma\epsilon_{j+1}}\right), \tag{6.1}$$

we can rewrite the proposed update equation as

$$S(t_{j+1}) = \frac{1}{2}\left\{S(t_j)\exp\left[\left(r-\frac{\sigma^2}{2}\right)\Delta t + \sigma\sqrt{\Delta t}\,\epsilon_{j+1}\right] + L'(t_{j+1})\right\}. \tag{6.2}$$

This formula is the average of two contributions: the standard stochastic Monte Carlo

update, Eq. (5.6), and the additional fitted term $L'(t_{j+1})$. The latter acts as a weight influencing the evolution of the asset price over time. We still allow random fluctuations, but the simulated price tends to follow the log-periodic behavior. The possibility of multiple trajectories remains; however, the paths now incorporate the expected trend.

Figures 28–31 show simulated trajectories obtained from Eq. (6.2). The values used to construct the functions $L'(t)$ derive from the fitting parameters in Table 3.

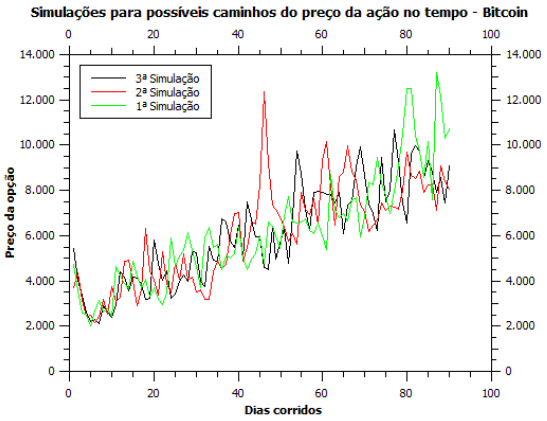


Figure 28. Bitcoin trajectories produced with the log-periodic correction.

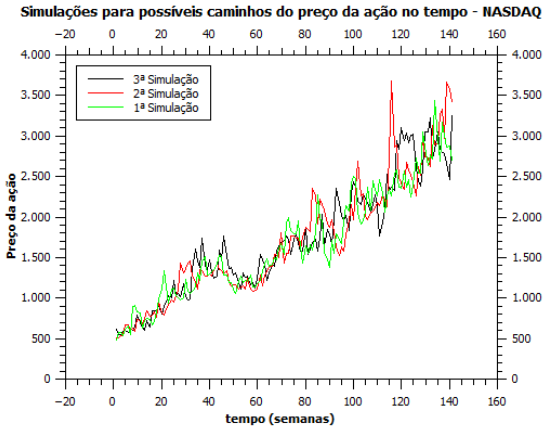


Figure 29. NASDAQ trajectories produced with the log-periodic correction.

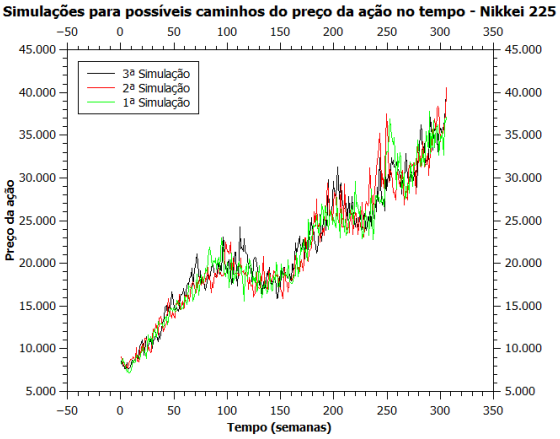


Figure 30. Nikkei 225 trajectories produced with the log-periodic correction.

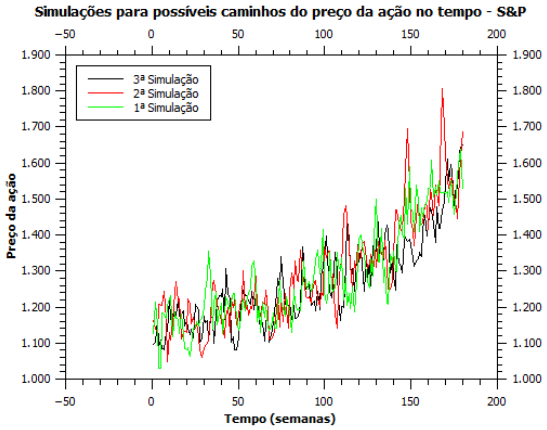


Figure 31. S&P 500 trajectories produced with the log-periodic correction.

The numerical results displayed in these figures were promising within the goals of the study. Stock-price randomness is retained, despite the increased prominence of the upward trend. Historical volatility also plays a major role in the proposed expression. Assets with lower volatility exhibit more pronounced log-periodic behavior, as in Figures 29 and 30, whereas assets with greater volatility tend to show stronger stochastic fluctuations.

For Bitcoin in Figure 28, the simulation uses $\sigma = 62.52\%$. Even though an upward trend is visible, the individual prices fluctuate markedly from one point to the next. This is consistent with the study's expectations. For a European option, the price at expiration is particularly important because it determines the payoff in Eq. (5.7). We

therefore require sufficient variation in terminal stock prices to represent possible real-world outcomes. Figure 28 shows such variation, and a larger simulation sample should produce a more representative range of terminal outcomes.

6.0.1 Changes to the Pricing Program

The modified algorithm retains much of the structure of the original program. Equation (5.6) is replaced by Eq. (6.1), and the fitted function $L(t)$ for the selected asset is defined at the beginning of the simulation. The parameters describing the log-periodic curve must be specified separately for each case. At the end of the chosen period, the algorithm again computes the payoff given by Eq. (5.7). Following n simulations, the mean payoff, multiplied by the appropriate adjustment factor, is used as the estimated option price.

6.1 Application to Bitcoin

We apply the proposed method to Bitcoin using the fitted parameters in Table 3 to define the log-periodic component of Eq. (6.1). For the example, we choose a historical volatility of 30%, approximately half the volatility estimated for December 14, 2017. The simulation inputs are given in Table 8.

Table 6.1: Parameters used in the new Bitcoin valuation example.

Variable	Value
Strike λ	\$10,000
Initial price S_0	\$4,614
Volatility σ	0.30
Interest rate r	5.35%
Days to expiration J	90

The resulting option price was \$218.78, which the original thesis reports as \$377.55 below the price obtained in the earlier Monte Carlo comparison (Table 6). An additional observation follows from comparing the observed price at expiration with the strike. The actual Bitcoin price at expiration was reported as \$11,595, while the strike was \$10,000. The difference is therefore \$1,595. In the previous comparison, the reported difference between the actual price and the strike was \$2,583. A smaller difference between strike and market price suggests that the option might have had a greater chance of being relevant at exercise.

Under the assumptions of the proposed model, choosing a strike informed by the predicted log-periodic trajectory can reduce both the gap between strike and observed expiration price and the estimated option premium. A buyer pays less for the option, while the intended reduction in extreme profit-and-loss outcomes may benefit the other side of the contract. As another illustration, keeping the other parameters unchanged but setting the strike to \$9,600—closer to the value suggested by the fit—produced an option

price of $101.08. Thus, the closer the selected strike lies to the model's predicted price in this example, the lower the option premium. In principle, the approach could help writers offer options at lower premiums if the fitted curve is a good forecast, though that premise would need independent validation.

We also explored how the estimated option price depends on historical volatility. Figure 32 shows the fitted behavior. The polynomial used for this analysis has the form $a+bx+cx^2$, with its reported maximum at $x = 84.697$ and coefficients displayed in the original chart. These results indicate that the new pricing rule depends strongly on volatility. As volatility rises, the paths may deviate more widely from the fitted trend, and the resulting option premium increases with greater uncertainty. Rather than contradicting the model's assumptions, this feature is consistent with its emphasis on the stochastic component. Figure 33 further illustrates the role of historical volatility.

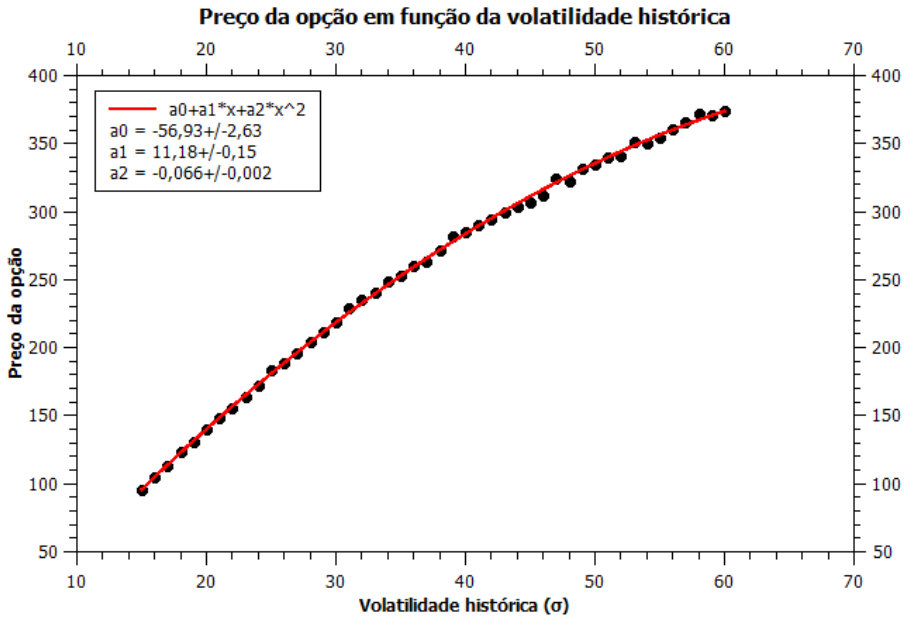


Figure 6.1: Estimated option price as a function of historical volatility.

An additional result is illustrated in Figure 34, which plots the option price against the strike for different volatilities. In this approach the strike can be informed by the log-periodic fit. Once the strike passes the level associated with the predicted expiration price, option premiums rise rapidly.

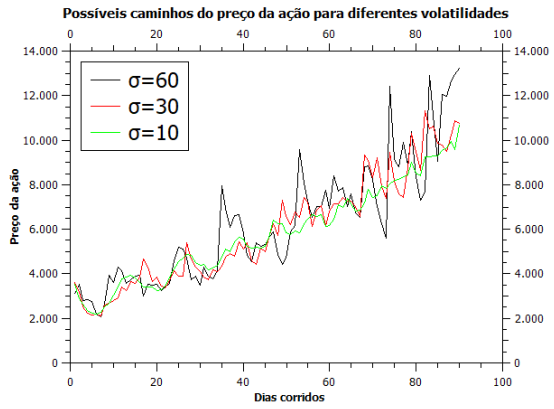


Figure 33. Possible underlying-price paths over time for different volatility assumptions.

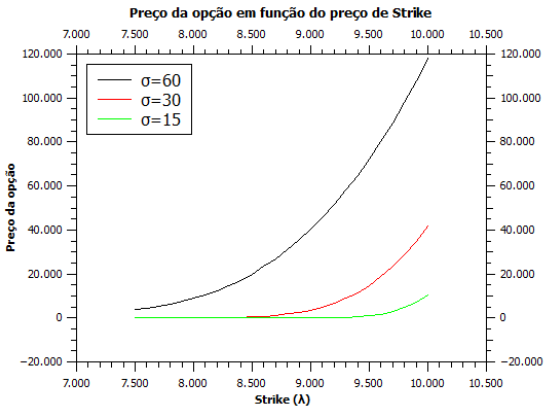


Figure 34. Option premium as a function of strike for different historical volatilities; displayed prices scaled by a factor of 1,000.

6.1.1 Considerations Regarding the Proposed Method

Figure 33 shows that the smaller the volatility, the more dominant the log-periodic trajectory becomes. We should consider what this implies. Low historical volatility indicates a system with relatively small price changes and consequently smaller returns; such a system may be more amenable to forecasting. High volatility is associated with larger price movements, which generally make forecasts less reliable. If we predict that a system will exhibit log-periodic behavior over a certain interval, a low-volatility system may offer a better chance that the forecast will be approximately realized. The program behaved in a way that was consistent with these assumptions. Because the revised model requires only limited changes to the earlier program, the convergence considerations from Chapter 5 remain relevant, especially the improvement in estimation quality as the number of paths n increases.

Forecasting market behavior is inherently difficult, and at some point a proposed future trajectory may be little more than an informed guess. Even so, the method remains internally coherent as an exploratory scenario exercise: it still simulates many possible paths of the underlying price (Figures 28–31) and therefore retains uncertainty about the terminal price. Accordingly, even a hypothesis that a stock will rise throughout a given interval can give the proposed modification a role as a scenario-based pricing experiment.

Conclusion

The results obtained in this study were satisfactory within the objectives originally proposed. Extensive reading was necessary to become better acquainted with econophysics, a fascinating field that seeks to describe phenomena as complex as the behavior of economic agents. We sought to maintain a quantitative perspective, recognizing that broader analyses outside the chosen scope could otherwise be inadequately developed. The investigations carried out, supported by a substantial bibliography, yielded interesting results and helped establish a theoretical foundation for the topics examined.

We also acknowledge that the analyses could be strengthened by a more extensive theoretical treatment of economics, which might explain additional perspectives on the behavior of the market participants discussed here. The proposed correction to the Monte Carlo pricing method worked well in the numerical experiments and generated promising results. The original work suggests that, with small or perhaps no modifications, the idea might be of interest to option writers who can anticipate, or at least hypothesize, future stock-price behavior. A possible real-world application could begin with an asset exposed to speculation and then reproduce the steps presented in this study. These observations remain exploratory, and practical use would require further testing and scrutiny of the model's financial assumptions.

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Appendix A

Original C-Language Monte Carlo Pricing Algorithm

The algorithm reproduced in the following pages was developed in the C programming language, drawing primarily on Crack (2009). A number of the results reported throughout the monograph were obtained by modifying this baseline implementation. Because the source code is part of the historical research artifact, its listing is preserved visually as it appeared in the Portuguese original, including original identifier names, code formatting, console prompts, and comments. Code strings should not be translated automatically if exact reproducibility is desired.

The four original appendix pages follow below at their original visual layout; the explanatory prose above is translated into English.

ANEXO A – Algoritmo da precificação por método de Monte Carlo

O algoritmo apresentado abaixo foi desenvolvido em *C*, e baseado nas instruções presentes principalmente no livro (CRACK, 2009). Diversos resultados apresentados durante o trabalho foram obtidos a partir de modificações nesse programa base.

```
#include <stdio.h>
#include <stdlib.h>
#include <time.h>
#include <math.h>
#define PI 3.141592654
#define NSUM 25

double gaussrand()
{
double x = 0;
int i;
for(i = 0; i < NSUM; i++)
x += (double)rand() / RAND_MAX;
x -= NSUM / 2.0;
x /= sqrt(NSUM / 12.0);

return x;
}

int main()

{
double st0;
double st_j;
double rndnmb;
float strike_t;
float sigma;
float rate;
double deltat;
double stock;
```

```
double arg;
double exparg;
double summ;
double totalsum;
double valorpay;
double estim;
double arg_n;
double arg1;
double arg2;

    FILE *result;
        char url[]="resultado.txt";
        result = fopen(url, "w");


int np, ip, sp;
int days;

srand(time(NULL)); /* gera semente aleatoria */

printf("------------Precificacao de Opcoes------------");
printf("\nDesenvolvido por Henrique Alves\n");

printf("\nInsira o preco da Acao:\n");
scanf("%lf", &stock);
printf("\nAcao: %f\n", stock);

printf("\nInsira o valor do Strike:\n");
scanf("%f", &strike_t);
printf("\nStrike: %f\n", strike_t);

printf("\nInsira o valor da taxa de juros:\n");
scanf("%f", &rate);
rate/=100;
printf("\nTaxa de juros: %f \n", rate);

printf("\nInsira o valor da taxa de volatilidade:\n");
scanf("%f", &sigma);
```

```
sigma/=100;
printf("\nTaxa de volatilidade: %f \n", sigma);

printf("\nInsira a quantidade de dias ate o vencimento:\n");
scanf("%d", &days);
printf("\nDias ate o vencimento: %d \n", days);

printf("\n");
printf("\n");

deltat = 0.002739726027f;
np=0;
arg_n = -(0.167123287f*rate);

for(sp=1; sp<=1000000; sp++)

{
for(ip=0; ip<days; ip++)

{
if(ip==0)

{
st0 = stock;
}

else
{
}

rndnmb = gaussrand();

arg1 = (rate-0.5*(sigma*sigma))*deltat;

arg2 = (sigma*0.05234239226*rndnmb);

arg = arg1+arg2;

exparg = (pow(M_E,arg)); //Exponencial do argumento
```

```
st_j = st0*(exparg);

st0 = st_j;
if(sp==1)
{
fprintf(result, "%0.3f \n", st_j);
}
}

np+=1;

if(strike_t-st_j > 0)

{

summ = strike_t-st0;

totalsum += summ;

}
else
{

}

}

printf("soma total = %f , sp = %i \n", totalsum, sp);

valorpay = totalsum/np;

estim = pow(M_E,arg_n)*valorpay;

printf("valor da opcao = %f\n", estim);


}
```